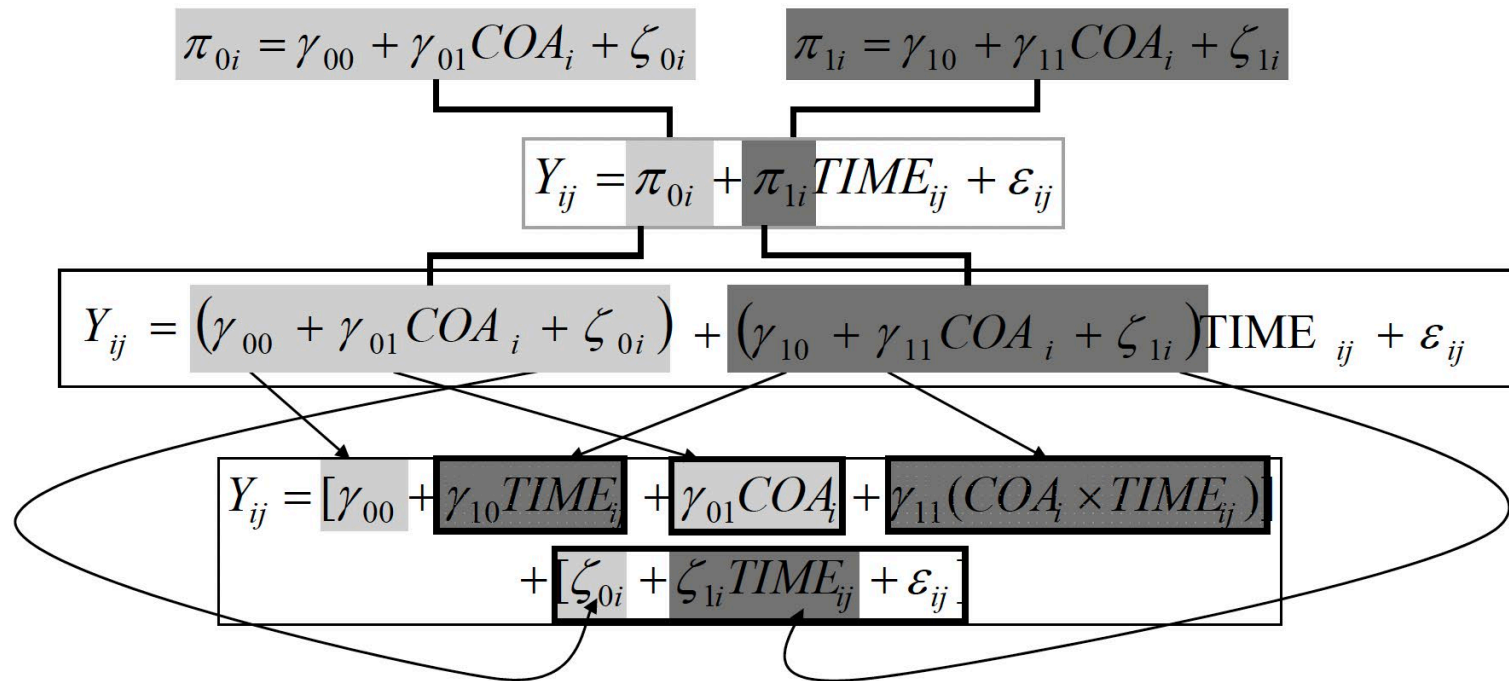


Lecture 5: Linear Growth Models

by Dr. Cook

Developing the composite specification of the multilevel model for change by substituting the level-2 submodels into the level-1 individual growth model



The composite specification shows how the outcome depends simultaneously on:

- ▣ *the level-1 predictor TIME and the level-2 predictor COA as well as*
- ▣ *the cross-level interaction, COA*TIME. This tells us that the effect of one predictor (TIME) differs by the levels of another predictor (COA)*

The composite specification also:

- ▣ *Demonstrates the complexity of the composite residual—this is not regular OLS regression*
- ▣ *Is the specification used by most software packages for multilevel modeling*
- ▣ *Is the specification that maps most easily onto the person-period data set... ⇒⇒⇒*

Structural part of level-2 model

(Singer and Willet pages 81-82; slides 30-31 Lecture 4)

$$\begin{aligned} Y_{ij} &= \pi_{0i} + \pi_{1i}TIME_{ij} + \varepsilon_{ij} \\ &= (\gamma_{00} + \gamma_{01}COA_i + \zeta_{0i}) + (\gamma_{10} + \gamma_{11}COA_i + \zeta_{1i})TIME_{ij} + \varepsilon_{ij}. \end{aligned}$$

Composite multilevel model for change

$$\begin{aligned} Y_{ij} &= [\gamma_{00} + \gamma_{10}TIME_{ij} + \gamma_{01}COA_i + \gamma_{11}(COA_i \times TIME_{ij})] \\ &\quad + [\zeta_{0i} + \zeta_{1i}TIME_{ij} + \varepsilon_{ij}], \end{aligned}$$

- Identifies the cross-level interaction
- More closely aligns with statistical programming
- **The composite and multilevel models are identical mathematically and logically**

Structural part of level-2 model

(Singer and Willet pages 81-82)

Level – 1
$$Y_{ij} = \pi_{0i} + \pi_{1i}TIME_{ij} + \varepsilon_{ij}$$

Level – 2
$$\pi_{0i} = \gamma_{00} + \gamma_{01}COA_i + \zeta_{0i}$$
$$\pi_{1i} = \gamma_{10} + \gamma_{11}COA_i + \zeta_{1i}$$

- First part about individual change (within-person change)
- Second part about interindividual change (between-person change)

From Last Lecture...

- Level-2 individual model of change (interindividual change in trajectories)
 - Define level-2 interindividual change model
 - Structural component
 - Stochastic component
- Estimating fixed effects
 - What are fixed effects?
 - Interpreting fixed effects
- Composite model (level-1 and level-2)
 - Fixed effects
 - Random effects

Learning Objectives

- Step 1: Fitting the unconditional models
 - Unconditional mean model
 - Unconditional growth model
- Step 2: Model building to the “final model”
 - Predictor only model
 - Predictor with control variable model
 - “Final” model
- Administration Stuff (Assignment # 2)
- Class exercise introduction: Friendship attachment and depression

Know before you go...

- ***Be sure you've examined empirical growth plots and fitted OLS trajectories.*** *You don't want to begin data analysis without being reasonably confident that you have a sound level-1 model*
- ***Be sure your person-period data set is correct.***
 - *Run simple diagnostics in whatever general purpose program you're comfortable with*
 - *Once again, you don't want to invest too much data analytic effort in a mis-formed data set*
- ***Don't jump in by fitting a range of models with substantive predictors.*** *Yes, you want to know "the answer," but first you need to understand how the data behave, so instead you should...*

First steps: Two unconditional models

1. ***Unconditional means model***—a model with no predictors at either level, which will help partition the total outcome variation
2. ***Unconditional growth model***—a model with TIME as the only level-1 predictor and no substantive predictors at level 2, which will help evaluate the baseline amount of change.

What these unconditional models tell us:

1. ***Whether there is systematic variation in the outcome worth exploring*** and, if so, where that variation lies (within or between people)
2. ***How much total variation there is both within- and between-persons***, which provides a baseline for evaluating the success of subsequent model building (that includes substantive predictors)

Step 1: Fitting the unconditional models

(Singer and Willett pgs. 92-104)

Unconditional mean model

- This model has one fixed effect
- This model examines the grand mean – the mean across all time points and observations
- We can interpret the sample mean
- We are really interested in examining the random effects (within- and between-person variance) to understand if we should move forward with modelling or not.

Unconditional multilevel models

- 2 Models – Quantify outcome variation
 - **Unconditional mean model** – No time consideration
 - **Unconditional growth model** – Across individual AND time
- 1. Is there any variation?
- 2. If so, where is the variation (within, between, or both)?
- 3. Provides a baseline for comparison!**

Unconditional mean model

$$Y_{ij} = \pi_{0i} + \varepsilon_{ij} \quad \text{OR} \quad Y_{ij} = \gamma_{00} + \zeta_{0i} + \epsilon_{ij}$$
$$\pi_{0i} = \gamma_{00} + \zeta_{0i}$$

- γ_{00} = grand mean (fixed effect)
 - ζ_{0i} = the amount person i's mean deviates from the population mean (between-person)
 - ϵ_{ij} = the amount the response on occasion j deviates from person i's mean (within-person)
 - $\epsilon_{ij} \sim N(0, \sigma_{\epsilon}^2)$
 - $\zeta_{0i} \sim N(0, \sigma_0^2)$
- ζ_{0i} = the distance between person i's mean and the population average mean
- The unconditional means represent the observed value of **Y** for individual **i** on occasion **j**, **which** is composed of deviations about the person-specific and the grand mean

Unconditional mean model

```
* Model A
xtmixed alcuse || id: , variance mle
```

```
Mixed-effects ML regression
Group variable: id
```

```
Number of obs      =      246
Number of groups   =       82
```

```
Obs per group: min =        3
                  avg =       3.0
                  max =        3
```

```
Wald chi2(0)       =        .
Prob > chi2        =        .
```

```
Log likelihood = -335.07799
```

alcuse	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
-----+-----						
_cons	.9219549	.0957073	9.63	0.000	.734372	1.109538
-----+-----						

Random-effects Parameters		Estimate	Std. Err.	[95% Conf. Interval]	
-----+-----					
id: Identity	var(_cons)	.5638619	.1191124	.3727018	.8530686
	var(Residual)	.561747	.0620346	.4524194	.6974936
-----+-----					

```
LR test vs. linear regression: chibar2(01) =    57.07 Prob >= chibar2 = 0.0000
```

The mean alcohol use for the average adolescent aged 14-16 is .92 (.85 when squared).

- **Between- person** variance different from zero. This suggests significant between-person variation
- **Within- person** variance different from zero. This suggests significant within-person variation

(use single parameter hypothesis test [see section 3.6 of book])

Overall,

- **The average adolescents alcohol use varies over time and adolescents differ from each other in their use**

Unconditional mean model

```
. estat icc
```

Intraclass correlation

$$\rho = \frac{\sigma_0^2}{\sigma_0^2 + \sigma_\varepsilon^2} \quad \hat{\rho} = \frac{0.564}{0.564 + 0.562} = 0.50$$

Level	ICC	Std. Err.	[95% Conf. Interval]	
id	.5009396	.0636977	.3785697	.6231971

- The proportion of the outcome variance that lies between people
 - Half of the total variation in alcohol use is attributable to differences between adolescents
- The ICC also calculates the amount of autocorrelation in the model.
 - There is a large amount of autocorrelation of the error term

The Unconditional Means Model (Model A)

Partitioning total outcome variation between and within persons

Level-1 Model: $Y_{ij} = \pi_{0i} + \varepsilon_{ij}$, where $\varepsilon_{ij} \sim N(0, \sigma_{\varepsilon}^2)$

Level-2 Model: $\pi_{0i} = \gamma_{00} + \zeta_{0i}$, where $\zeta_{0i} \sim N(0, \sigma_0^2)$

Composite Model:

$$Y_{ij} = \gamma_{00} + \zeta_{0i} + \varepsilon_{ij}$$

Within-person deviations

Grand mean across individuals and occasions

Person-specific means

		Parameter	Model A
Fixed Effects			
Initial status, π_{0i}	Intercept	γ_{00}	0.922*** (0.096)
Variance Components			
Level 1	Within-person	σ_{ε}^2	0.562*** (0.062)
Level 2	In initial status	σ_0^2	0.564*** (0.119)

Within-person variance

Between-person variance

Let's look more closely at these variances....

Using the unconditional means model to estimate the Intraclass Correlation Coefficient (ICC or ρ)

*Major purpose of the unconditional means model:
To partition the variation in Y into two components*

*Quantifies the
amount of variation
between individuals
regardless of time*

$$\hat{\sigma}_0^2 = 0.564^{***}$$

*Estimated between-
person variance*

$$\hat{\sigma}_\varepsilon^2 = 0.562^{***}$$

*Estimated within-
person variance*

*Quantifies the
amount of variation
within individuals
over time*

*Intraclass correlation compares
the relative magnitude of these
VCs by **estimating the
proportion of total
variation in Y that lies
“between” people***

$$\rho = \frac{\sigma_0^2}{\sigma_0^2 + \sigma_\varepsilon^2}$$

$$\hat{\rho} = \frac{0.564}{0.564 + 0.562} = 0.50$$

*An estimated **50% of the
total variation in alcohol
use is attributable to
differences between
adolescents***

Unconditional growth model

- This model allows us to examine the growth parameters by person and time.
- The fixed effects are the intercept and slope.
- The random effects are composed of the variance components or error terms.

Unconditional growth model

$$Y_{ij} = \pi_{0i} + \pi_{1i}TIME_{ij} + \epsilon_{ij}$$

$$\pi_{0i} = \gamma_{00} + \zeta_{0i}$$

$$\pi_{1i} = \gamma_{10} + \zeta_{1i},$$

$$Y_{ij} = \gamma_{00} + \gamma_{10} * time_{ij} + \zeta_{0i} + \zeta_{1i} * time_{ij} + \epsilon_{ij}$$

- γ_{00} = average intercept (fixed effect)
- γ_{10} = average slope (fixed effect)
- ζ_{0i} = the amount person i's intercept deviates from the population intercept
- ζ_{1i} = the amount person i's slope deviates from the population slope
- ϵ_{ij} = the amount the response on occasion j deviates from person i's true change

trajectory

- $\epsilon_{ij} \sim N(0, \sigma_{\epsilon}^2)$

σ_{ϵ}^2 Summarizes the scatter of each persons data around his or her own linear change trajectory

- $\zeta_{0i} \sim N(0, \sigma_0^2)$

σ_0^2 Summarizes the between person variability

- $\zeta_{1i} \sim N(0, \sigma_1^2)$

σ_1^2 Summarizes the within person variability

- ζ_{0i} and ζ_{1i} have covariance σ_1^2

Unconditional growth model

*** Model B**

```
xtmixed alcuse age_14 || id: age_14 , cov(un) variance mle
```

```
min = 3
avg = 3.0
max = 3
```

```
Wald chi2(1) = 18.78
Prob > chi2 = 0.0000
```

Log likelihood = -318.30554

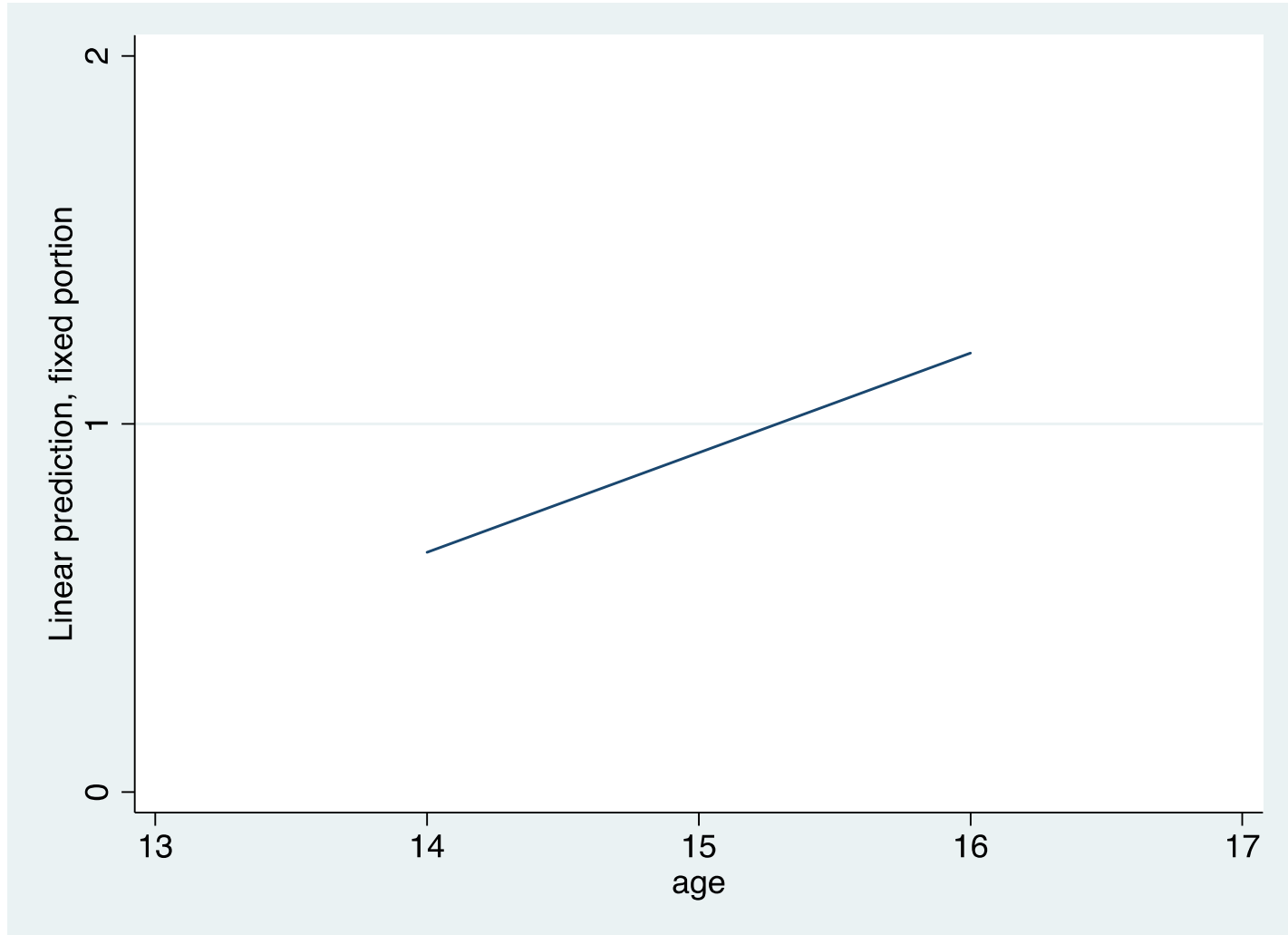
alcuse	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
age_14	.2706514	.0624547	4.33	0.000	.1482424	.3930604
_cons	.6513035	.1050803	6.20	0.000	.4453499	.8572571

Random-effects Parameters	Estimate	Std. Err.	[95% Conf. Interval]	
id: Unstructured				
var(age_14)	.1512026	.0564703	.0727206	.3143845
var(_cons)	.6243568	.1480618	.3922622	.9937777
cov(age_14,_cons)	-.0684396	.0700778	-.2057896	.0689105
var(Residual)	.3372922	.0526762	.2483539	.4580803

LR test vs. linear model: chi2(3) = 79.70

Prob > chi2 = 0.0000

Unconditional growth model



Unconditional growth model

```
#Model B
model.b <- lme(alcuse ~ age_14 , data=alcohol1, random= ~ age_14 | id, method="ML")
summary(model.b)
```

Linear mixed-effects model fit by maximum likelihood

Data: alcohol1

	AIC	BIC	logLik
	648.6111	669.643	-318.3055

Random effects:

Formula: ~age_14 | id

Structure: General positive-definite, Log-Cholesky parametrization

	StdDev	Corr
(Intercept)	0.7901617	(Intr)
age_14	0.3888470	-0.223
Residual	0.5807690	

Fixed effects: alcuse ~ age_14

	Value	Std.Error	DF	t-value	p-value
(Intercept)	0.6513035	0.10551005	163	6.172905	0
age_14	0.2706514	0.06271014	163	4.315912	0

Correlation:

(Intr)	
age_14	-0.441

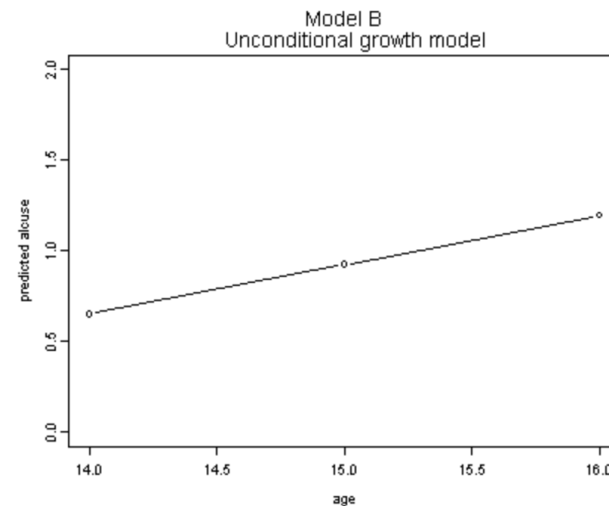
Standardized Within-Group Residuals:

	Min	Q1	Med	Q3	Max
	-2.47998885	-0.38401088	-0.07553389	0.39001356	2.50684931

Number of Observations: 246

Number of Groups: 82

```
fixef.b <- fixef(model.b)
fit.b <- fixef.b[[1]] + alcohol1$age_14[1:3]*fixef.b[[2]]
plot(alcohol1$age[1:3], fit.b, ylim=c(0, 2), type="b",
     ylab="predicted alcuse", xlab="age")
title("Model B \n Unconditional growth model")
```



Unconditional growth model

- The residual variance summarizes the average scatter of an individual's observed outcome values of alcohol use around his/her own true change trajectory.
- The decline from .562 to .337 suggests that 40% of the within-person variation in alcohol use is associated with a linear change in time.

Random-effects Parameters	Estimate	Std. Err.	[95% Conf. Interval]	
-----+-----				
id: Unstructured				
var(age_14)	.1512035	.0564703	.0727213	.3143849
var(_cons)	.6243576	.1480618	.3922629	.9937785
cov(age_14,_cons)	-.0684405	.0700779	-.2057906	.0689096
-----+-----				
var(Residual)	.3372916	.052676	.2483536	.4580793
-----+-----				

- Level-2 variances assess the scatter of a person's intercept and slope about the population average change trajectory
- Because these are statistically significant it means that there remains unexplained heterogeneity at level two that could be explained further by the introduction of additional predictors at this level

Unconditional growth model

. xtmixed, stddeviations

Random-effects Parameters	Estimate	Std. Err.	[95% Conf. Interval]	
id: Unstructured				
sd(age_14)	.3888478	.0726123	.2696675	.5607
sd(_cons)	.7901625	.0936907	.6263084	.996884
corr(age_14,_cons)	-.2227467	.1884036	-.5476977	.1605954
sd(Residual)	.5807686	.0453504	.4983512	.6768163



- Difference between the true intercept and true slope
- We conclude that the relationship between these two growth parameters is negative and weak and may not be different from zero

The Unconditional Growth Model (Model B)

A baseline model for change over time

Level-1 Model: $Y_{ij} = \pi_{0i} + \pi_{1i}TIME_{ij} + \varepsilon_{ij}$, where $\varepsilon_{ij} \sim N(0, \sigma_{\varepsilon}^2)$

Level-2 Model: $\pi_{0i} = \gamma_{00} + \zeta_{0i}$
 $\pi_{1i} = \gamma_{10} + \zeta_{1i}$ where $\begin{bmatrix} \zeta_{0i} \\ \zeta_{1i} \end{bmatrix} \sim N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_0^2 & \sigma_{01} \\ \sigma_{10} & \sigma_1^2 \end{bmatrix}\right)$

Composite Model:

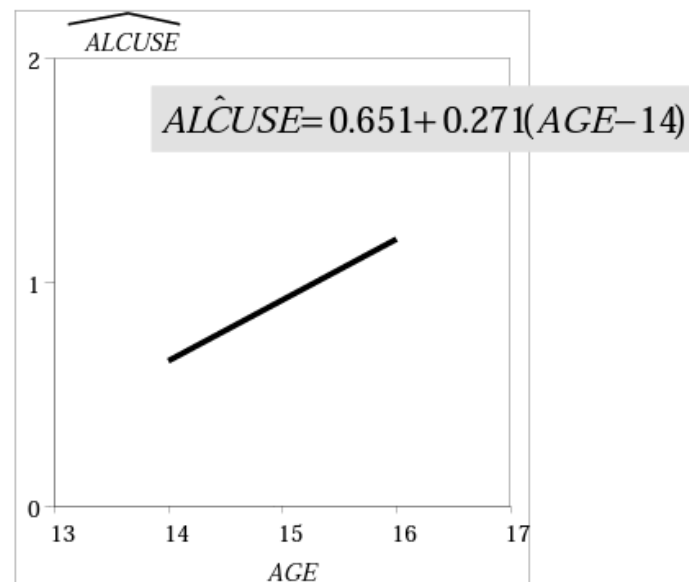
$$Y_{ij} = \gamma_{00} + \gamma_{10}TIME_{ij} + [\zeta_{0i} + \zeta_{1i}TIME_{ij} + \varepsilon_{ij}]$$

Composite residual

Average true initial status at AGE 14

Average true rate of change

	Parameter	Model B
Fixed Effects		
Initial status, π_{0i}	Intercept.....	γ_{00}
		0.651*** (0.105)
Rate of change, π_{1i}	Intercept	γ_{10}
		0.271*** (0.062)



		Parameter	Model B
Variance Components			
Level 1	Within-person	σ_ϵ^2	0.337*** (0.053)
Level 2	In initial status	σ_0^2	0.624*** (0.148)
	In rate of change	σ_1^2	0.151** (0.056)
	Covariance	σ_{01}	-0.068 (0.070)

Level-1 (within person)

There is still unexplained within-person residual variance

Level-2 (between-persons):

- *There is between-person **residual variance in initial status** (but careful, because initial status has changed)*
- *There is between-person **residual variance in rate of change** (should consider adding a level-2 predictor)*
- *Estimated res. covariance between **initial status and change** is n.s.*

So...what has been the effect of moving from an unconditional means model to an unconditional growth model?

		Parameter	Model B
Variance Components			
Level 1	Within-person	σ_ϵ^2	0.337*** (0.053)
Level 2	In initial status	σ_0^2	0.624*** (0.148)
	In rate of change	σ_1^2	0.151** (0.056)
	Covariance	σ_{01}	-0.068 (0.070)

Level-1 (within person)

There is still unexplained within-person residual variance

Level-2 (between-persons):

- *There is between-person **residual variance in initial status** (but careful, because initial status has changed)*
- *There is between-person **residual variance in rate of change** (should consider adding a level-2 predictor)*
- *Estimated res. covariance between **initial status and change** is n.s.*

- We have found that 40% of the within-person variation in alcohol use is associated with a linear change in time
- Half of total variance in alcohol use attributed to differences among young adults
- The within and between remain after the inclusion of time

So...what has been the effect of moving from an unconditional means model to an unconditional growth model?

Step 2: Model building to the “final model”

(Singer and Willett pgs. 104-113)

Where we've been and where we're going...

What these unconditional models tell us:

- 1. About half the total variation in ALCUSE is attributable to differences among teens*
- 2. About 40% of the within-teen variation in ALCUSE is explained by linear TIME*
- 3. There is significant variation in both initial status and rate of change— so it pays to explore substantive predictors (COA & PEER)*

How do we build statistical models?

- Use all your intuition and skill you bring from the cross sectional world*
 - Examine the effect of each predictor separately*
 - Prioritize the predictors,*
 - Focus on your “question” predictors*
 - Include interesting and important control predictors*
- Progress towards a “final model” whose interpretation addresses your research questions*

But because the data are longitudinal, we have some other options...

- 📌 Multiple level-2 outcomes (the individual growth parameters)—each can be related separately to predictors*
- 📌 Two kinds of effects being modeled:*
 - 📌 Fixed effects*
 - 📌 Variance components*
 - 🔴 Not all effects are required in every model*

Model building to the “final model”

- This is a mixture of art and science
- You want to make sure your model building is guided by theory, research, concepts, etc.
- You will run several models, but the models you place in the manuscript should be useful.
- I usually use the following models –
 - Unconditional mean model
 - Unconditional growth model
 - Conditional model with covariate(s) only
 - Conditional model with covariate(s) and main independent variable

Model Building: Main IV growth model

Mixed-effects ML regression
Group variable: **id**

Number of obs = 246
Number of groups = 82

Obs per group:

min = 3
avg = 3.0
max = 3

Wald chi2(3) = 35.77
Prob > chi2 = 0.0000

Log likelihood = -310.60131

alcuse	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
coa	.743212	.1945656	3.82	0.000	.3618704	1.124554
age_14	.2929552	.0842277	3.48	0.001	.1278719	.4580384
c.coa#c.age_14	-.0494299	.1253894	-0.39	0.693	-.2951887	.1963288
_cons	.3159517	.1306953	2.42	0.016	.0597936	.5721098

Estimated difference in
initial use for a child of
alcoholic and non
alcoholic parents

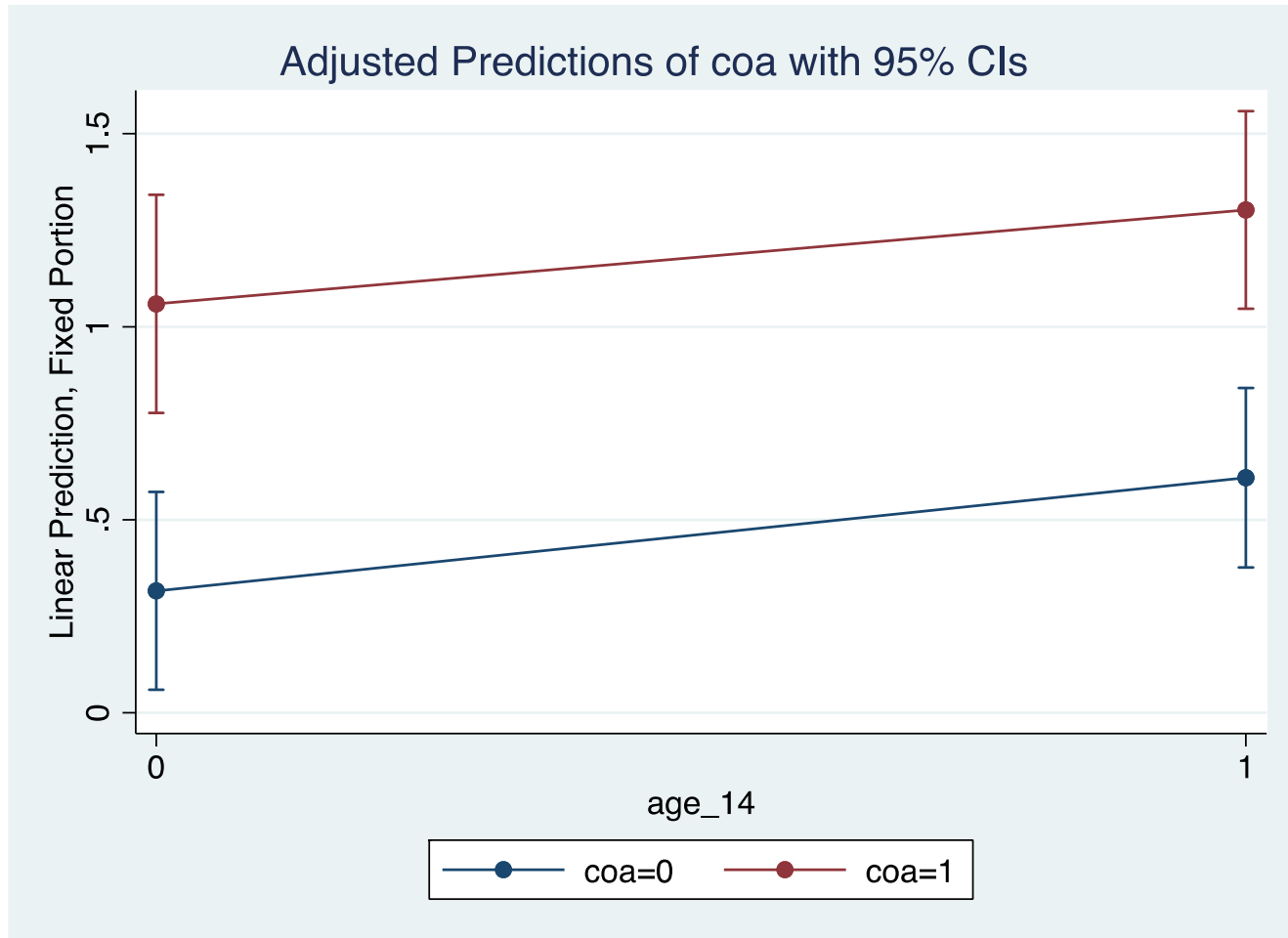
Estimated
rate of
change in
alcohol use
for a child
of non-
alcoholic
parents

Estimated initial use for
a child of non alcoholic
parents

The estimated differential in the rate of change in alcohol
use between children of alcoholics and non-alcoholics

Model Building: Main IV growth model

```
xtmixed alcuse i.coa age_14 i.coa#c.age_14 || id: age_14 , cov(un) variance mle  
margins coa, at(age_14=(0 1))  
marginsplot
```



What will our analytic strategy be?

Because our research interest focuses on the effect of COA, essentially treating PEER is a control, we're going to proceed as follows...



Model C: COA predicts both initial status and rate of change.

Level-1/level-2 specification

Model	level-1 model	level-2 model
C	$Y_{ij} = \pi_{0i} + \pi_{1i}TIME_{ij} + \varepsilon_{ij}$	$\pi_{0i} = \gamma_{00} + \gamma_{01}COA_i + \zeta_{0i}$ $\pi_{1i} = \gamma_{10} + \gamma_{11}COA_i + \zeta_{1i}$
D	$Y_{ij} = \pi_{0i} + \pi_{1i}TIME_{ij} + \varepsilon_{ij}$	$\pi_{0i} = \gamma_{00} + \gamma_{01}COA_i + \gamma_{02}PEER_i + \zeta_{0i}$ $\pi_{1i} = \gamma_{10} + \gamma_{11}COA_i + \gamma_{12}PEER_i + \zeta_{1i}$
E	$Y_{ij} = \pi_{0i} + \pi_{1i}TIME_{ij} + \varepsilon_{ij}$	$\pi_{0i} = \gamma_{00} + \gamma_{01}COA_i + \gamma_{02}PEER_i + \zeta_{0i}$ $\pi_{1i} = \gamma_{10} + \gamma_{12}PEER_i + \zeta_{1i}$

Model D: Adds PEER to both Level-2 sub-models in Model C.

Model E: Simplifies Model D by removing the non-significant effect of COA on change.

What will our analytic strategy be?

Because our research interest focuses on the effect of COA, essentially treating PEER is a control, we're going to proceed as follows...



Model C: COA predicts both initial status and rate of change.

Level-1/level-2 specification

Model	level-1 model	level-2 model
C	$Y_{ij} = \pi_{0i} + \pi_{1i}TIME_{ij} + \varepsilon_{ij}$	$\pi_{0i} = \gamma_{00} + \gamma_{01}COA_i + \zeta_{0i}$ $\pi_{1i} = \gamma_{10} + \gamma_{11}COA_i + \zeta_{1i}$
D	$Y_{ij} = \pi_{0i} + \pi_{1i}TIME_{ij} + \varepsilon_{ij}$	$\pi_{0i} = \gamma_{00} + \gamma_{01}COA_i + \gamma_{02}PEER_i + \zeta_{0i}$ $\pi_{1i} = \gamma_{10} + \gamma_{11}COA_i + \gamma_{12}PEER_i + \zeta_{1i}$
E	$Y_{ij} = \pi_{0i} + \pi_{1i}TIME_{ij} + \varepsilon_{ij}$	$\pi_{0i} = \gamma_{00} + \gamma_{01}COA_i + \gamma_{02}PEER_i + \zeta_{0i}$ $\pi_{1i} = \gamma_{10} + \gamma_{12}PEER_i + \zeta_{1i}$

Model D: Adds PEER to both Level-2 sub-models in Model C.

Model E: Simplifies Model D by removing the non-significant effect of COA on change.

Model C: Assessing the uncontrolled effects of COA (the question predictor)

		Parameter	Model B	Model C
Fixed Effects				
Initial status, π_{0i}	Intercept	γ_{00}	0.651*** (0.105)	0.316*** (0.131)
	COA	γ_{01}		0.743*** (0.195)
	PEER	γ_{02}		
Rate of change, π_{2i}	Intercept	γ_{10}	0.271*** (0.062)	0.293*** (0.084)
	COA	γ_{11}		-0.049 (0.125)
	PEER	γ_{12}		
Variance Components				
Level 1	Within-person	σ_{ϵ}^2	0.337*** (0.053)	0.337*** (0.053)
Level 2	In initial status	σ_0^2	0.624*** (0.148)	0.488** (0.128)
	In rate of change	σ_1^2	0.151** (0.056)	0.151* (0.056)
	Covariance	σ_{01}	-0.068 (0.070)	-0.059 (0.066)
Deviance			636.6	621.2
AIC			648.6	637.2
BIC			663.0	656.5

-p < .10; *p < .05; **p < .01; ***p < .001

Fixed effects

- Est. initial value of ALCUSE for non-COAs is 0.316 ($p < .001$)
- Est. differential in initial ALCUSE between COAs and non-COAs is 0.743 ($p < .001$)
- Est. annual rate of change in ALCUSE for non-COAs is 0.293 ($p < .001$)
- Estimated differential in annual rate of change between COAs and non-COAs is -0.049 (ns)

Variance components

- Within person VC is identical to B's because no predictors were added
- Initial status VC declines from B: COA "explains" 22% of variation in initial status (but still stat sig. suggesting need for level-2 pred's)
- Rate of change VC unchanged from B: COA "explains" no variation in change (but also still sig suggesting need for level-2 pred's)

Next step?

- Remove COA? Not yet—question predictor
- Add PEER—Yes, to examine controlled effects of COA

Model D: Assessing the controlled effects of COA (the question predictor)

		Parameter	Model C	Model D
Fixed Effects				
Initial status, π_{0i}	Intercept	γ_{00}	0.316*** (0.131)	-0.317*** (0.148)
	COA	γ_{01}	0.743*** (0.195)	0.579*** (0.162)
	PEER	γ_{02}		0.694*** (0.112)
Rate of change, π_{2i}	Intercept	γ_{10}	0.293*** (0.084)	0.429*** (0.114)
	COA	γ_{11}	-0.049 (0.125)	-0.014 (0.125)
	PEER	γ_{12}		-0.150~ (0.086)
Variance Components				
Level 1	Within-person	σ_{ϵ}^2	0.337*** (0.053)	0.337*** (0.053)
Level 2	In initial status	σ_0^2	0.488** (0.128)	0.241** (0.093)
	In rate of change	σ_1^2	0.151* (0.056)	0.139* (0.055)
	Covariance	σ_{01}	-0.059 (0.066)	-0.006 (0.055)
Deviance			621.2	588.7
AIC			637.2	608.7
BIC			656.5	632.8

~ $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .001$

Fixed effects of COA

- Est. diff in ALCUSE between COAs and non-COAs, controlling for PEER, is 0.579 ($p < .001$)
- No sig. Difference in rate of change

Fixed effects of PEER

- Teens whose peers drink more at 14 also drink more at 14 (initial status)
- Modest neg effect on rate of change ($p < .10$)

Variance components

- Within person VC unchanged (as expected)
- Still sig. variation in both initial status and change—need other level-2 predictors
- Taken together, PEER and COA explain
 - 61.4% of the variation in initial status
 - 7.9% of the variation in rates of change

Next step?

- If we had other predictors, we'd add them because the VCs are still significant
- Simplify the model? Since COA is not associated with rate of change, why not remove this term from the model?

Model E: Removing the non-significant effect of COA on rate of change

			Parameter	Model D	Model E	
Fixed Effects						
Initial status, π_{0i}	Intercept	γ_{00}	-0.317*** (0.148)	-0.314*** (0.146)	Fixed effects of COA <i>Controlling for PEER, the estimated diff in ALCUSE between COAs and non-COAs is 0.571 (p<.001)</i>	
	COA	γ_{01}	0.579*** (0.162)	0.571*** (0.146)		
	PEER	γ_{02}	0.694*** (0.112)	0.695*** (0.111)		
Rate of change, π_{2i}	Intercept	γ_{10}	0.429*** (0.114)	0.425*** (0.106)	Fixed effects of PEER <i>Controlling for COA, for each 1 pt difference in PEER, initial ALCUSE is 0.695 higher (p<.001) but rate of change in ALCUSE is 0.151 lower (p<.10)</i>	
	COA	γ_{11}	-0.014 (0.125)			
	PEER	γ_{12}	-0.150~ (0.086)	-0.151~ (0.085)		
Variance Components						
Level 1	Within-person	σ_{ϵ}^2	0.337*** (0.053)	0.337*** (0.053)	Variance components are unchanged suggesting little is lost by eliminating the main effect of COA on rate of change (although there is still level-2 variance left to be predicted by other variables)	
Level 2	In initial status	σ_0^2	0.241** (0.093)	0.241** (0.093)		
	In rate of change	σ_1^2	0.139* (0.055)	0.139* (0.055)		
	Covariance	σ_{01}	-0.006 (0.055)	-0.006 (0.055)	Partial covariance is indistinguishable from 0. <i>After controlling for PEER and COA, initial status and rate of change are unrelated</i>	
			Deviance	588.7	588.7	
			AIC	608.7	606.7	
			BIC	632.8	628.4	

~ $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .001$

Where we've been and where we're going...

- *Let's call Model E our tentative "final model" (based on not just these results but many other analyses not shown here)*
- *Controlling for the effects of PEER, the estimated differential in ALCUSE between COAs and nonCOAs is 0.571 ($p < .001$)*
- *Controlling for the effects of COA, for each 1-pt difference in PEER: the average initial ALCUSE is 0.695 higher ($p < .001$) and average rate of change is 0.151 lower ($p < .10$)*

Displaying prototypical trajectories

Recentering predictors to improve interpretation

*Alternative strategies for hypothesis testing:
Comparing models using Deviance statistics and information criteria*

Additional comments about estimation

Summary of Findings

- Adolescent alcohol usage differs depending on parental history of alcoholism as well as the amount of alcohol peers consumed initially.
- At each level of peer drinking, children of non-alcoholics had less alcohol use than children of alcoholics
- In addition, regardless of parental drinking status, if peers drank at age 14, adolescent alcohol use increased (see pgs. 111-112 of book to calculate by hand)

Administrative Stuff

Assignment # 2: Prob #2

Longitudinal Analysis of Public Health Data Rubric for Assignment 2

ASSESSMENT CATEGORIES (POINTS)	POINTS/NOTES	SCORE
Part 1 (2-3 sentences or less) - An additional predictor selected and operationalized - Time variable is defined		.5 points
Part 2 (2-3 sentences or less) - Level-1 and Level-2 equations correctly written - Level-1 and Level-2 equations correctly defined - The composite equation is correctly written - The composite equation is correctly defined		1 point
Part 3 (2-3 sentences or less) - Descriptive statistics properly obtained for “wide” form - Descriptive statistics properly discussed for “wide” form - Profile plot(s) properly obtained for “wide” form - Profile plot(s) properly discussed for “wide” form		1.5 points
Part 4 (4-5 sentences or less) - Empirical growth plot(s) obtained and defined - Non-parametric standardization approach utilized and interpreted - Parametric standardization approach utilized and interpreted - Individual OLS regressions conducted and visualized with the mean trajectory line. This graph is also interpreted		2 points
Part 5 (350 words or less) - Sample means for intercept and slope obtained and interpreted - Sample variances for intercept and slope obtained and interpreted - Correlation between intercept and slope obtained and interpreted - Initial thoughts concerning the intercept and slope of the level one model and the level 2 model - Clearly stated and informed by descriptive statistics outlined in assignment		4 points
STYLE/PRESENTATION/ACADEMIC RIGOR - Writing is focused and well-organized – quality is very good to excellent - No grammatical or spelling errors - Written in scientific language		1 point
		TOTAL (10 points)